

Hong Yuan Cao

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EDUCATION

Boston University

Boston, MA

Double Major: Bachelor of Arts in Computer Science and Economics, GPA: 3.77/4.0 Expected Graduation: May 2026

Relevant Courses: Distributed Systems, Databases, Software Engineering, Analysis of Algorithms, Computer Systems, Probability in Computing, Combinatoric Structures, Geometric Algorithms, Discrete Mathematics, Data Structures

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C#, Go, SQL, HTML/CSS, LaTeX, Assembly

Databases & Cloud: MySQL, PostgreSQL, Redis, AWS (Lambda, S3, CloudFront, DynamoDB), Alibaba Cloud RDS

Frameworks & Libraries: Django, React.js, Vue.js, React Native, Flask, Node.js, Spring Boot, JUnit, pytest, PyTorch, NumPy, Pandas, Scikit-learn, NLTK, Matplotlib, Seaborn, BeautifulSoup, Axios, Terraform, Docker, Git

EXPERIENCES

Software Engineering Intern

Shanghai, China

Tospur Real Estate Consulting Co., Ltd

June 2024 – August 2024

- Implemented a centralized data warehouse with **Django** and **Alibaba Cloud RDS**, creating a scalable backend system that integrated 4 internal **MySQL** databases and reduced manual data aggregation time by **45%**
- Implemented **RESTful APIs** with Django REST Framework that connected the database to frontend client applications, enabling **200+** consultants to access and update project records in real-time
- Optimized database performance on Alibaba Cloud RDS by implementing **B-tree indexing** on high-traffic query columns and configuring **connection pooling** with optimal timeout settings, reducing query latency by **23%**

Machine Learning Intern

Shanghai, China

Anban Tech

May 2024 – June 2024

- Preprocessed and managed multi-modal data for a LLM designed for automated test code generation, leveraging **Pandas** and **regex** to clean, tokenize, and structure raw code
- Curated a **2.2GB** training dataset of **JUnit 5** test cases from MIT/Apache-licensed repositories, using **Git API** for extraction and storing in **JSON** format with proper attribution metadata
- Developed validation scripts in **Python** with senior engineers to identify and standardize code formatting inconsistencies across the dataset, successfully normalizing **95%** of test cases for improved model training

Software Engineering Intern

San Francisco, CA

Tonic – Sprout Technologies INC

June 2023 – October 2023

- Implemented responsive UI components and animations using **React.js**, increasing landing page engagement by **24%**
- Built a social media crawler using **Python (BeautifulSoup4)** that identified 200+ qualified influencers to target
- Processed 3,000+ media contacts using **Python (Pandas)** and Excel, improving outreach response rates by **25%**

PROJECTS

Serverless Image Processing Pipeline | AWS S3, Lambda, CloudFront, DynamoDB, Terraform, React

- Provisioned serverless image processing infrastructure with **Terraform**, utilizing **S3**, **Lambda**, and **CloudFront** to reduce image delivery latency by **38%**
- Implemented **Python**-based **Lambda** functions triggered by **S3** events to automatically resize uploaded images
- Built **React.js** frontend connecting to AWS services through **API Gateway**, storing metadata in **DynamoDB**

Multi-Currency Budget Tracking Mobile App | Django, React Native, PostgreSQL, Android Studio

- Developed a expense tracking app with **React Native**, featuring currency conversion via third-party API integration
- Built RESTful APIs with **Django REST Framework** and **PostgreSQL**, implementing **composite indexing** on user-specific data fields and optimizing **ORM queries** to reduce data retrieval time by **65%**
- Engineered offline data persistence using **AsyncStorage** local caching with **JWT** authentication, reducing API requests by **57%** and improving average user interaction response time from **1.2s to 400ms**

Movie Recommendation Platform | Flask, Vue.js, Python, Javascript, PostgreSQL

- Integrated movie database API with **Flask** to create an interactive dashboard using **Vue.js** components
- Built content-based recommendation algorithm using **CountVectorizer** and **cosine similarity** to analyze metadata
- Created secure user authentication with **Flask-Login** and **bcrypt**, storing user data in **PostgreSQL**

Neural Language Model Implementation | Python, PyTorch

- Implemented transformer architecture in **PyTorch** with token and positional embeddings for sequential prediction
- Designed custom attention mechanisms for next-token prediction, enabling model to generate coherent text
- Implemented training pipeline with **AdamW** optimizer and batch processing, reducing model perplexity by 30%